

Agile Delivery Plan

Multi-Property Rental Management System

1. Project Delivery Strategy

Methodology

- Agile Scrum
- 2-week sprints
- Continuous integration & deployment pipeline from Sprint 3 onward
- Every sprint ends with:
 - Demo build
 - Stakeholder review
 - Backlog refinement

Primary User Roles

- Landlord / Portfolio Owner
 - Property Manager
 - Finance Team
 - Tenant
 - System (Automations)
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2. High Level System Architecture

Frontend

- React / Next.js Web Dashboard
- Responsive Tenant Portal

Backend

- REST API (Node.js or Python)
- PostgreSQL multi-tenant database

Core Integrations (Across Sprints)

- M-Pesa API
 - Stripe API
 - SMS Gateway (Africa's Talking / Twilio)
 - WhatsApp Business API
 - AWS S3 (document storage)
 - Email Service (SendGrid / AWS SES)
 - Cron / Background Job Worker
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3. Definition of Done (Global)

A sprint is complete when:

- Feature deployed to staging
 - Unit + integration tests pass
 - API documented
 - UX approved
 - Security reviewed
 - Demo delivered
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4. Sprint Roadmap Overview

Sprint	Theme
Sprint 0	Project foundation & architecture
Sprint 1	Authentication & RBAC
Sprint 2	Organization & Property Management
Sprint 3	Units & Tenant Core Profiles
Sprint 4	Lease Engine & Deposit Logic
Sprint 5	Financial Ledger Engine
Sprint 6	Automated Billing & Background Jobs
Sprint 7	Payment Integrations & Reconciliation
Sprint 8	Expenses & P&L Reporting

Sprint 9 Dashboards & Analytics

Sprint 10 Tenant Portal

Sprint 11 Notifications & Communication

Sprint 12 Document Management

Sprint 13 Audit Logs & Security Hardening

Sprint 14 Performance & Scalability

Sprint 15 UAT & Production Launch

SPRINT 0 — Project Foundation & Architecture

Goal

Create the technical backbone of the system.

Deliverables

DevOps & Infrastructure

- Repository setup
- Branching strategy (GitFlow)
- CI/CD pipeline (GitHub Actions)
- Environment configs (Dev / Staging / Prod)
- Docker setup

Backend Setup

- Project structure
- Database connection
- Migration tool (Alembic / Prisma)
- Base API framework
- Health check endpoint

Frontend Setup

- React.js app scaffold
- UI framework setup

- Global layout
- Authentication placeholders

Database Foundation

Create core tables:

- organizations
- users
- roles
- permissions
- audit_logs (initial)

Integrations

- Email service setup
 - AWS S3 bucket creation
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SPRINT 1 — Authentication & RBAC

Goal

Secure login and role-based system access.

Deliverables

User Authentication

- Registration
- Login / Logout
- Password reset
- JWT session management
- Refresh tokens

RBAC System

Roles:

- Landlord
- Property Manager
- Finance
- Tenant

Permissions:

- Role-based route guards
- Property access scoping

Security

- Password hashing
- Rate limiting
- Basic audit logging

Integrations

- Email service (password reset)
 - JWT / OAuth library
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SPRINT 2 — Organization & Property Management

Goal

Enable multi-tenant architecture and property isolation.

Deliverables

Organization Module

- Create organization
- Invite users
- Assign roles per organization

Property Module

- Create property
- Property settings
- Assign property managers
- Property context switcher backend logic

UI

- Property switch dropdown
- Property list page

Integrations

- None external (internal architecture sprint)
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SPRINT 3 — Units & Tenant Profiles

Goal

Implement tenant onboarding lifecycle.

Deliverables

Unit Management

- Add/edit/delete units
- Unit status:
 - Vacant
 - Occupied
 - Maintenance

Tenant Profiles

Fields:

- Name
- National ID
- Phone
- Email
- Income source
- Next of Kin (mandatory)

Search Engine

Global search by:

- Name
- Phone
- ID
- Unit number

Integrations

- PII encryption library

SPRINT 4 — Lease Engine & Deposit State Machine

Goal

Implement lease lifecycle and deposit holding logic.

Deliverables

Lease Management

- Create lease
- Assign tenant to unit
- Lease status:
 - Active
 - Notice given
 - Inspection
 - Final settlement
 - Closed

Deposit Engine

- Record security deposit
- Lock deposit funds
- Prevent counting as revenue

Integrations

- None (core financial logic)
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SPRINT 5 — Financial Ledger Engine

Goal

Build the accounting core.

Deliverables

Transaction System

Transaction types:

- Debit
- Credit
- Refund
- Adjustment

Suspense Account

- Unmatched payments holding area

Ledger Rules

- Double entry ready structure
 - Per property isolation
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SPRINT 6 — Automated Billing & Cron Jobs

Goal

Automate monthly invoicing.

Deliverables

Billing Engine

- Monthly rent generator
- Late fee rules
- Arrears tracking

Background Jobs

- Cron job scheduler
- Job retry logic
- Queue system

Integrations

- Background worker (Redis / RabbitMQ)

SPRINT 7 — Payment Integrations & Reconciliation

Goal

Automate rent collection.

Deliverables

M-Pesa Integration

- STK Push
- Webhook receiver
- Payment validation

Stripe Integration

- Card payments
- Webhook handling

Auto Reconciliation Flow

1. Receive webhook
2. Match reference → tenant
3. Auto credit ledger
4. Send receipt notification

Integrations

- M-Pesa API
 - Stripe API
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SPRINT 8 — Expenses & P&L Reporting

Goal

Track property costs and profitability.

Deliverables

Expense Tracking

- Upload receipts
- Categorize expenses
- Property expense ledger

Financial Reports

- Net Operating Income
- Monthly P&L
- Expense breakdown charts

Integrations

- AWS S3 (receipt storage)
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SPRINT 9 — Dashboards & Analytics

Goal

Executive visibility.

Deliverables

Portfolio Dashboard

- Total revenue
- Occupancy rate
- Arrears
- NOI

Property Dashboard

- Property P&L
 - Vacancy trends
 - Payment trends
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SPRINT 10 — Tenant Portal

Goal

Self-service tenant experience.

Deliverables

Tenant Login

- Tenant authentication
- View statement
- Download statement PDF

Real-Time Balance View

- Payment history
 - Rent due alerts
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SPRINT 11 — Notifications & Communication

Goal

Automate communication.

Deliverables

Messaging Engine

- Rent due SMS
- Payment received SMS
- Email receipts
- Broadcast announcements

Integrations

- SMS Gateway
- WhatsApp Business API

SPRINT 12 — Document Management

Goal

Centralized legal document storage.

Deliverables

- Lease upload
- ID upload
- Receipt storage
- Document permissions

Integrations

- AWS S3
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SPRINT 13 — Audit Logs & Security Hardening

Deliverables

- Full audit trail
 - PII encryption at rest
 - Role penetration testing
 - Security review
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SPRINT 14 — Performance & Scalability

Deliverables

- Load testing
- Query optimization

- Caching layer (Redis)
 - CDN setup
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SPRINT 15 — UAT & Production Launch

Deliverables

- User acceptance testing
 - Bug fixing sprint
 - Production deployment
 - Monitoring & alerts
 - Documentation & training
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End of Agile Execution Plan